Created in Cambodia, November 2012 (Revised August 2018)

Rice Blast

Recognize the problem

Rice blast starts as a tiny red spot on the leaf or stem and may infect the whole plant. A typical blast lesion on a rice leaf is grey at the centre, has a dark border, and is diamond-shaped or spindle-shaped (large in the middle and tapering toward both ends) (Fig. 1). When there is too much rice blast, the plant cannot make food from sunlight and it will eventually die. The loss is even higher when rice nodes or rice panicles are infected and break after sometime (Fig. 2).

Fig 1. Spindle-shaped lesions (Photo by IRRI)



Blast lesion on node. (Photo by www.IRRI.org)



Background

Rice blast is caused by a fungus and is present wherever rice is cultivated. Areas which are generally wet and with cool temperature in daytime are more favourable to blast disease. This applies particularly to upland and rain-fed rice field. Rice can have blast in all growth stages. However, leaf blast incidence tends to lessen as plants mature and develop adult plant resistance to the disease.

Management

- Use resistant variety if available.
- Apply a balanced amount of plant nutrients, especially Nitrogen.
- Ensure good drainage of flooded fields and nurseries.
- Remove rice straw and weed hosts (e.g. Echinochloa spp) from the field. Dry the field during the fallow period to kill the fungus and reduce the amount of inoculum in the soil and plant residues.
- Using chemicals (fungicides) such as edifanphos, isoprothiolane, tricyclazole and mancozeb to kill the blast were found effective against leaf and neck blasts. Spray when lesions start to appear and continue spraying 3-7 days later.

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > ***Pyricularia grisea***

The recommendations in this factsheet are relevant to: Cambodia

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